

TRPA1 natural antagonist provides therapeutics for MRSA infection through HIF-1a and MAPK modulation

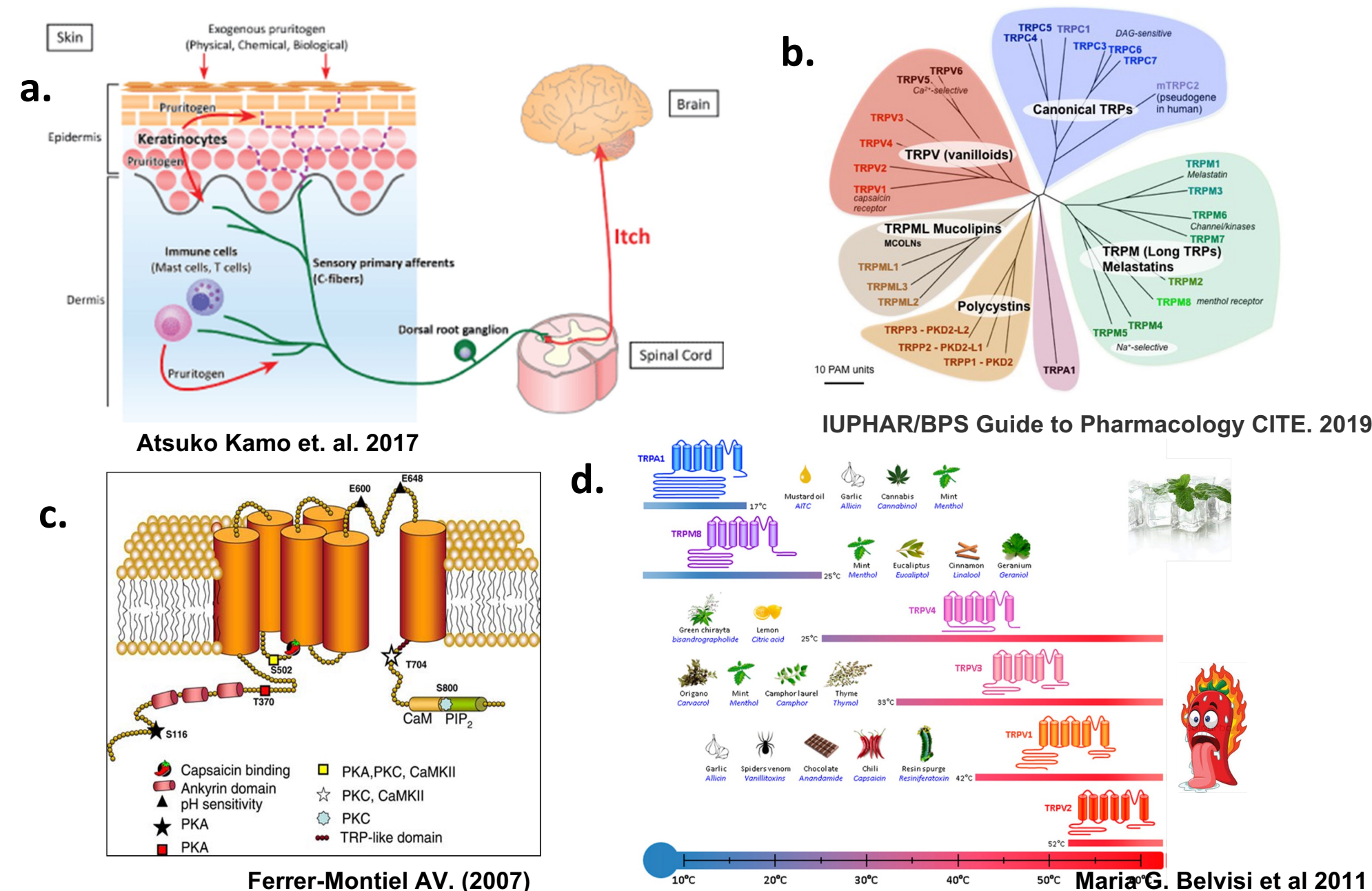
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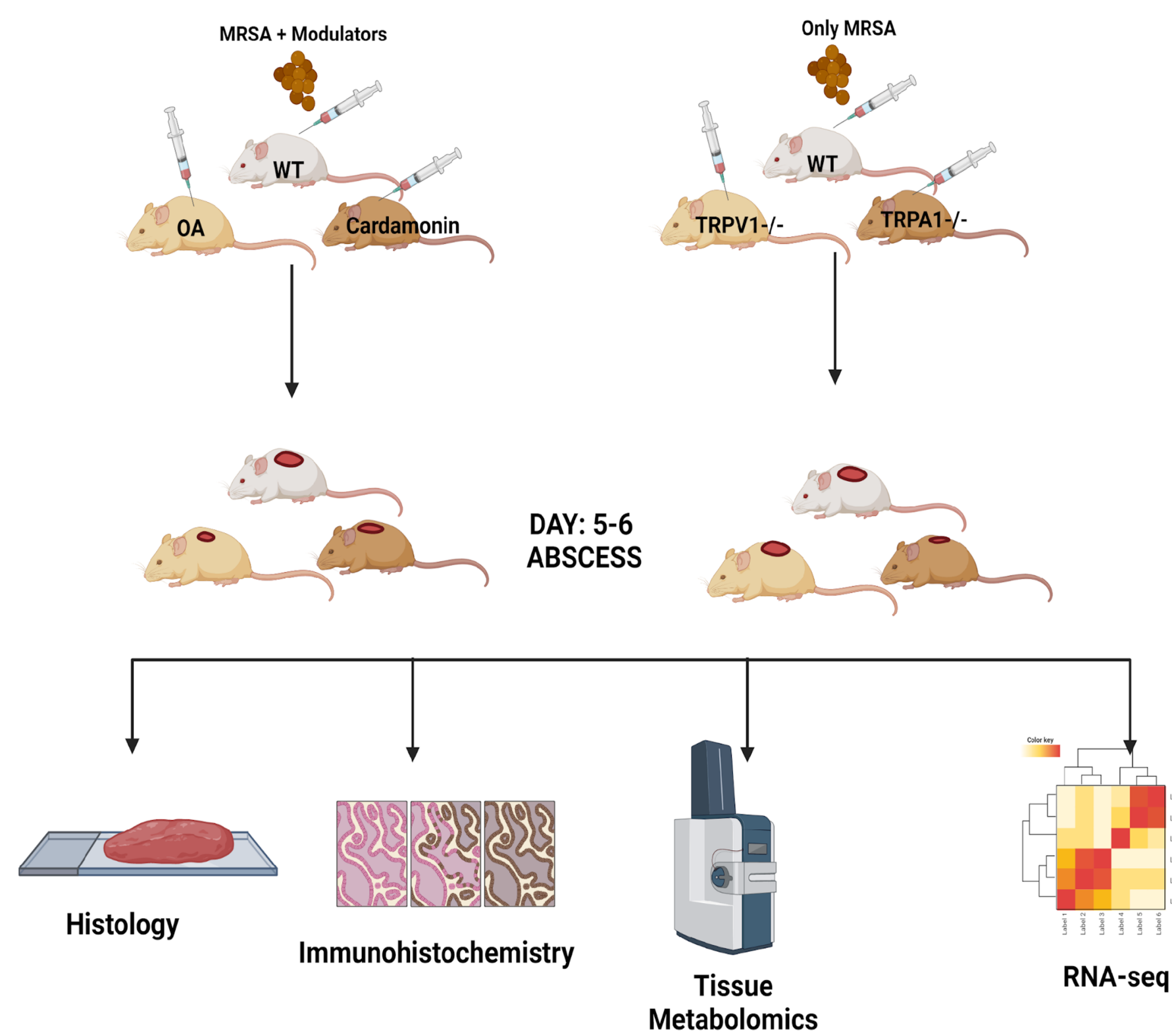
Abstract: Due to a surge in antibiotic resistance, skin infection and diseases caused by Methicillin-resistant *Staphylococcus aureus* (MRSA) is a major public health burden. Furthermore, antibiotic-resistance demands new treatment approaches for MRSA infection such as novel antimicrobial agents and host defense peptides. The thermo-ion channels Transient Receptor Potential (TRP-) A1 and V1 have been identified as modulators of *S. aureus* quorum sensing in cell culture models, however, their effects on in vivo infection control are unknown. In this study, we have investigated the therapeutic role of TRP ion channel on MRSA skin infection through TRPV1 and TRPA1 knockout mice and natural inhibitors. While TRPV1 modulation did not show much effect on infection, TRPA1^{-/-} mice and those treated with cardamonin (a natural inhibitor of TRPA1) demonstrated significantly reduced infection severity and abscess size. Tissue transcriptomic data coupled with immunohistochemistry revealed that TRPA1 inhibition impacted heat shock protein expression (HSP), which further modulated the HIF-1a and MAPK pathways and reduced IL4 expression. Tissue metabolomics reveals that TRP modulation impacts purine and glycosaminoglycan pathways. Multi-omics analysis reveals that diacylglycerol metabolism was the likely bridge between metabolic and immunologic impacts. Our findings suggest that natural TRPA1 antagonists provide a promising, and cost-effective therapeutic approach for reducing MRSA infection and present a novel underlying molecular mechanism. By shedding light on the potential of TRP ion channel modulation, our study contributes to ongoing efforts to address the global health challenge posed by antibiotic-resistant *S. aureus*.

Background: TRP ion channel



TRP ion channel family. a.) Shows schematic representation of itch response. b.) Shows phylogenetic tree of TRP ion channel superfamily members. c.) Shows monomeric structure of TRPV1 ion channel. d.) Shows thermosensitive ion channel along with their natural modulators.

Methodology



Schematic representation of the methodology and the downstream methods used in order to carry out this study.

Results

Figure 1:TRP-knockout and modulation reduce staph infection

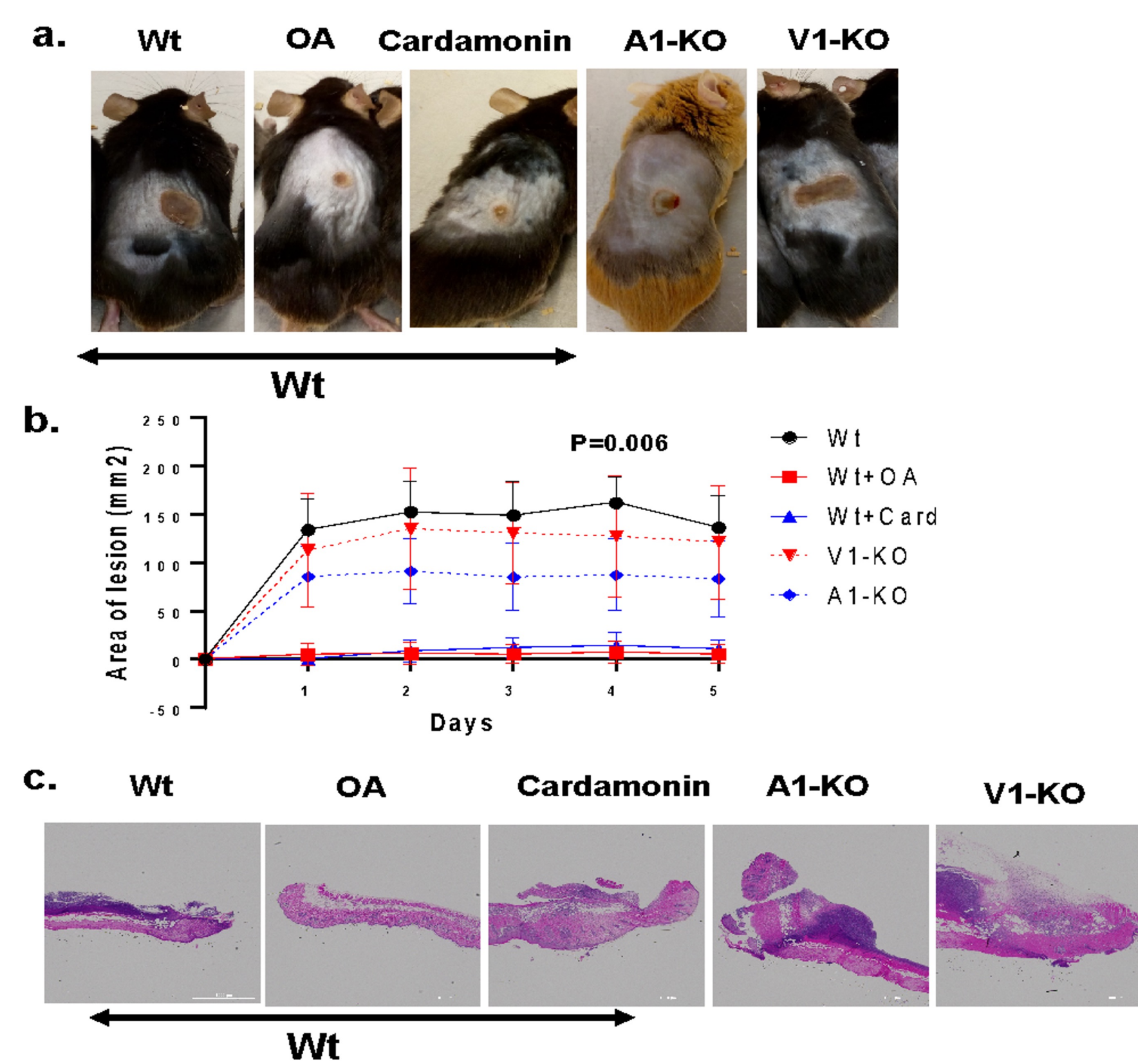


Fig.1: TRPA1-KO mice inhibits the MRSA infection. a.) Representative mice images after subcutaneous injection of MRSA and TRP modulators. Each mice treatment group had replicates n=3-5 mice. Image shown are mean of replicates from group b.) Shows quantification of lesion size in each treatment group. c.) Shows H and E-stained images from each group shows increased neutrophil.

Figure 2: TRP KO mice have reduced IL-4 and IL-17 level

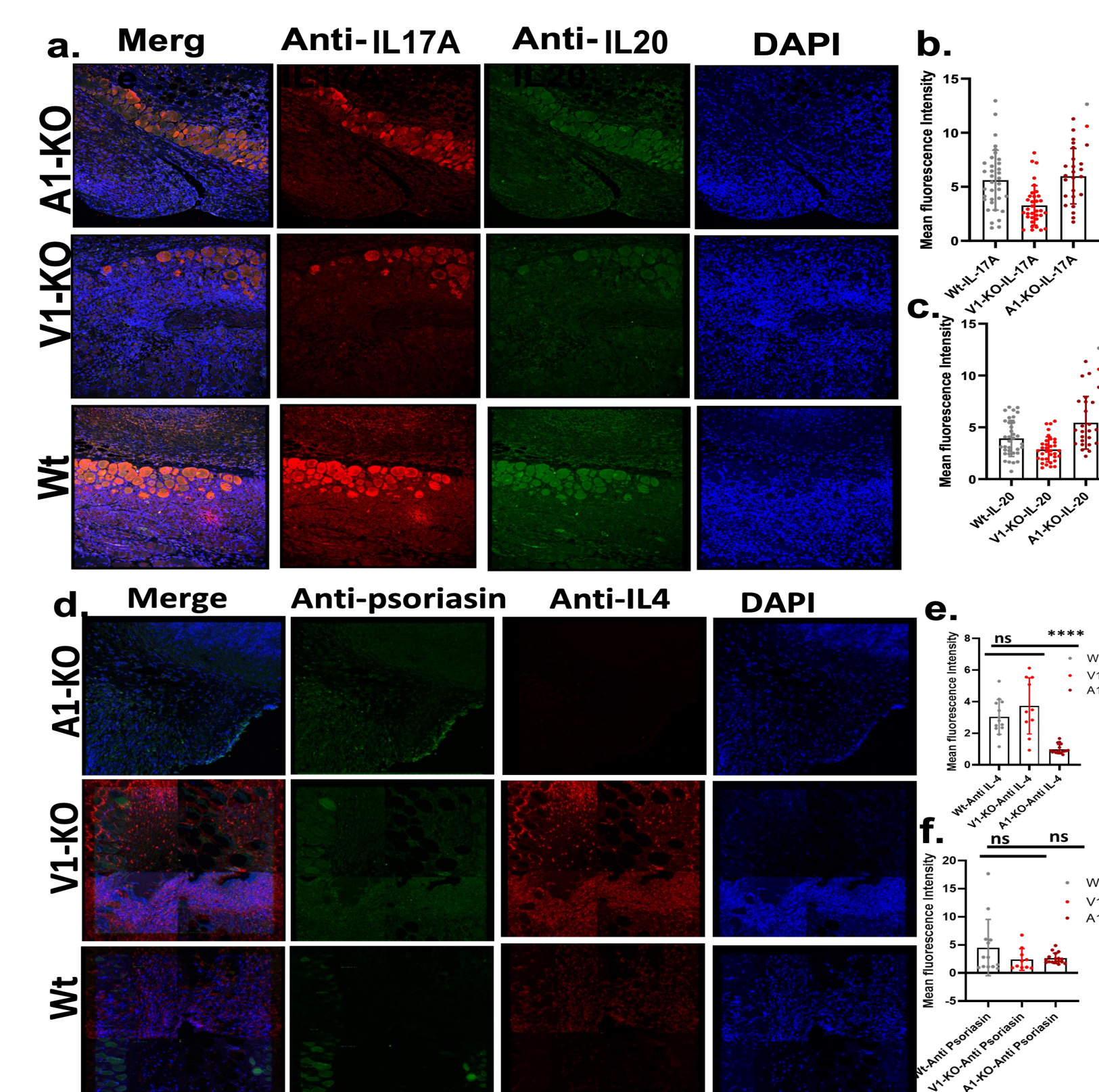


Fig. 2: Immunohistochemistry (IHC) of the skin tissue. a.) Shows images stained with IL17A, IL-20 and DAPI in each mice tissue group. b and c.) Shows quantification of IL17A and IL20. d.) Shows confocal images of Psoriasis, IL-4, and DAPI stained mice tissue e and f.) Shows quantification of IL-4 and Psoriasis respectively.

Figure 3: RNA seq analysis identifies TRPA1 modulation impacts HIF-1a and MAPK pathways

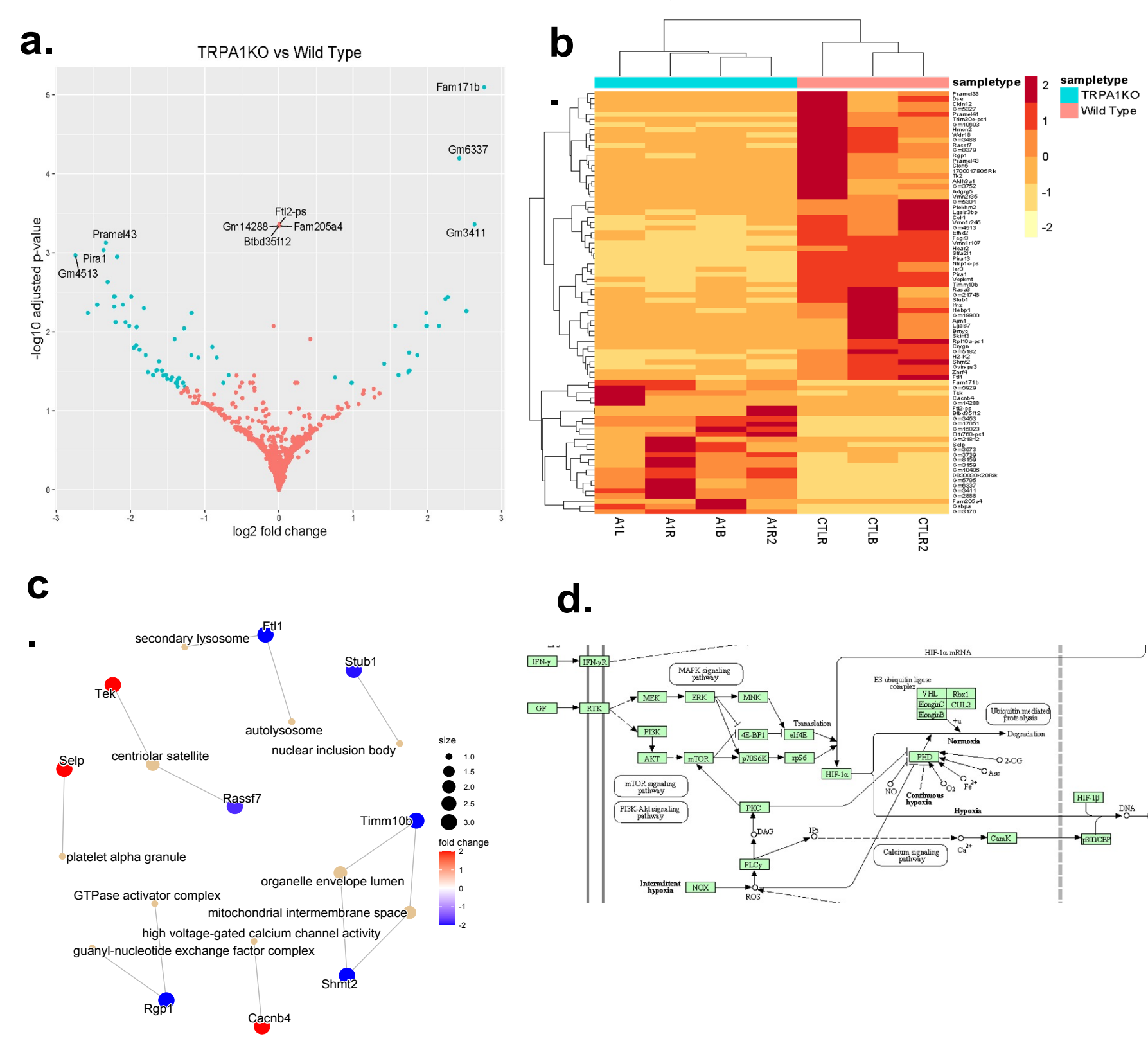


Fig.3: RNA seq results: Volcano plot and heatmap showing the differentially expressed genes between TRPA1KO and wild type (a). And the expression of differential expressed genes in each animal (b). category netplot shows the relationships between the genes associated with the top ten most significant GO terms and the fold changes of the significant genes associated with these terms (color)(c). TRPA1-KO reveals that HIF1a and MAPK pathways where impacted in MRSA infection (d).

Figure 4: RNA seq analysis reveals fewer TRPV1 differentially expressed transcripts

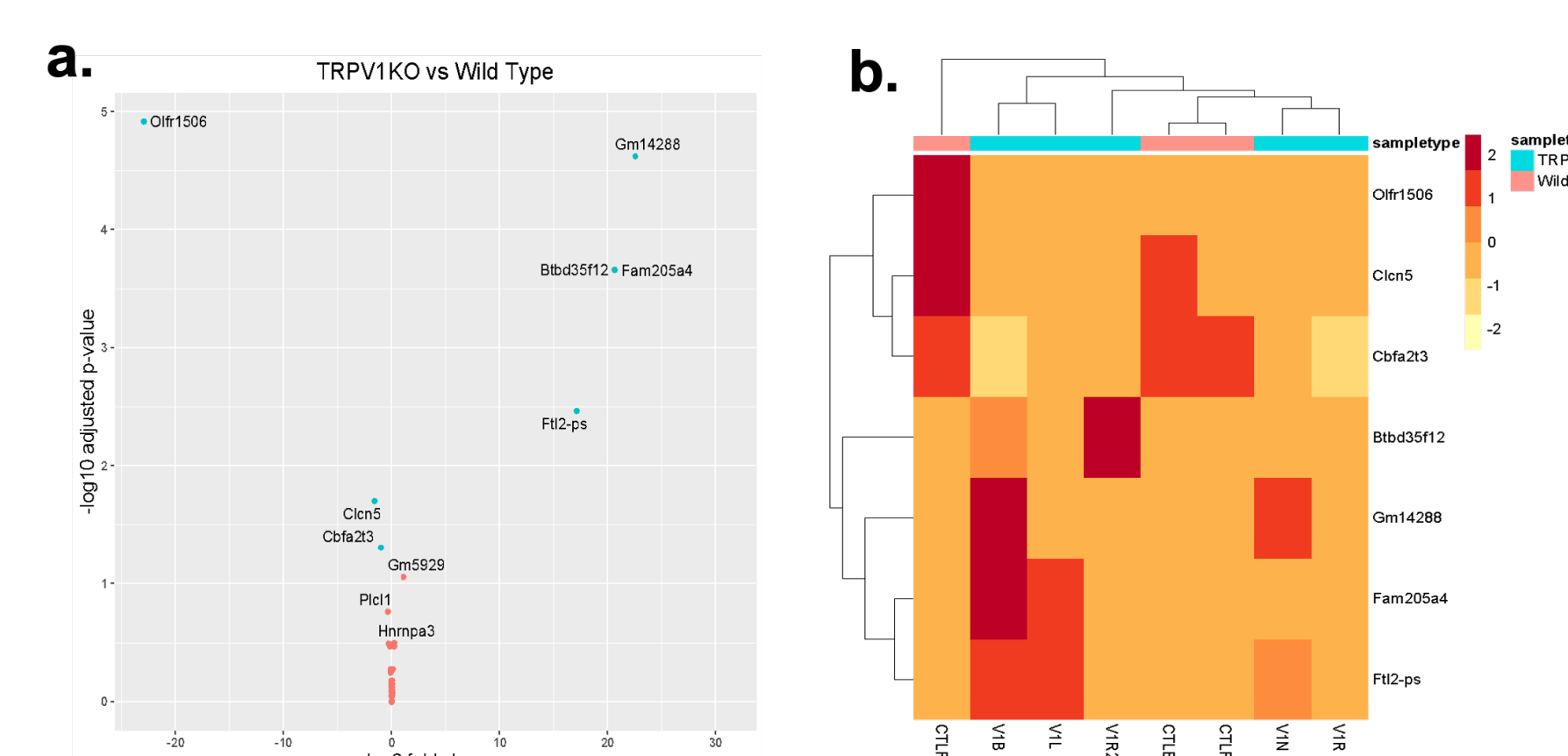


Fig.4: RNA seq results: Volcano plot and heatmap showing the differentially expressed genes between TRPV1KO and wild type (a). And the expression of differential expressed genes in each animal (b).

Figure 5: MRSA infected TRP-wt/antagonist/ TRPA1-KO mice shows differential metabolites:

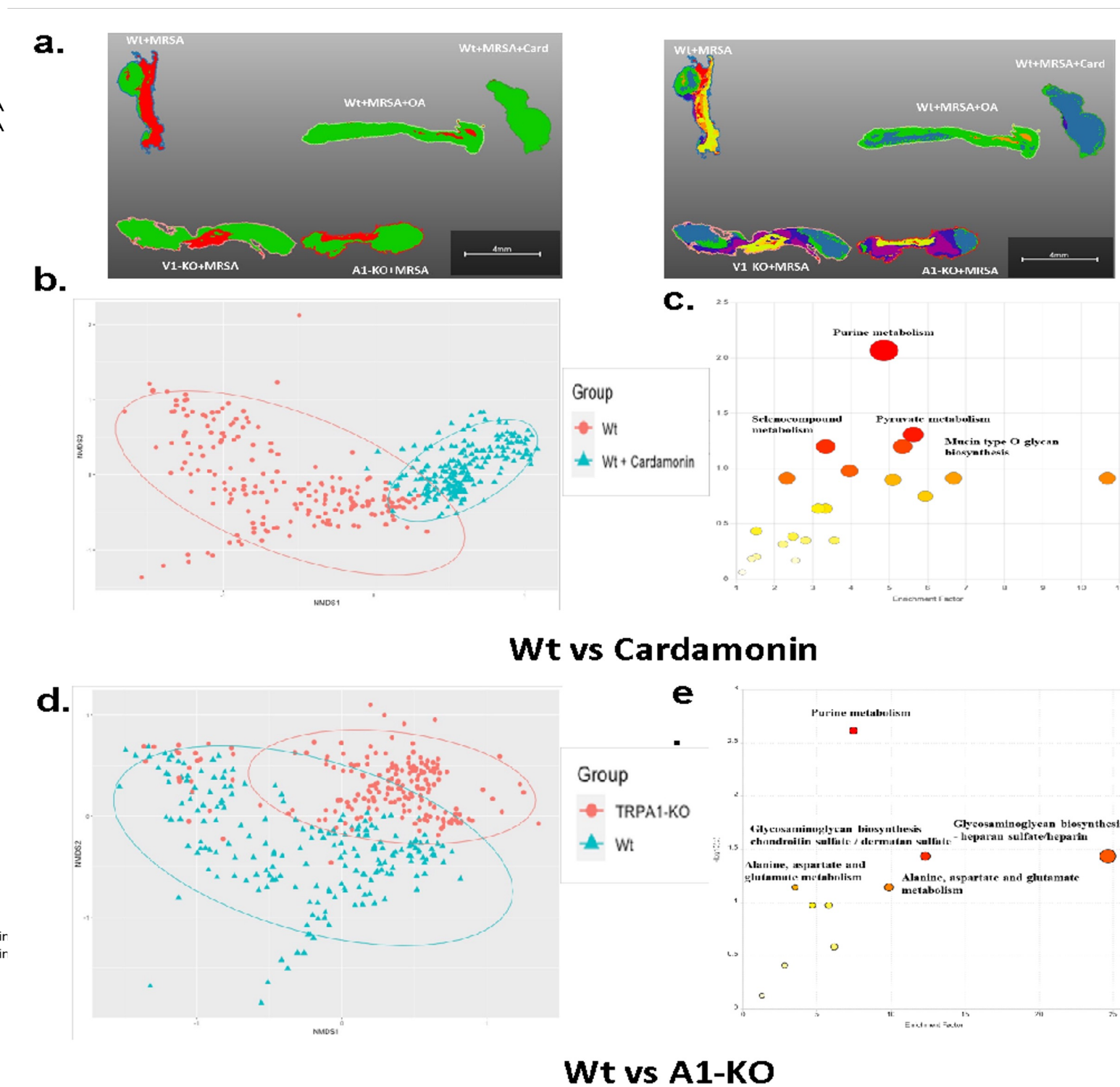


Fig. 5: MALDI imaging of MRSA infected mice tissue segmentation (a). NMDS plot showing the metabolomic profile difference between the Wt and natural inhibitor (b) and Wt and TRPA1KO (d).Metabolic pathway plots shows which pathways are significantly altered when comparing Wt and Natural inhibitor (c) and Wt and TRPA1KO (e).

Figure 6: Multi Omics analysis shows DAG and phospholipids were impacted

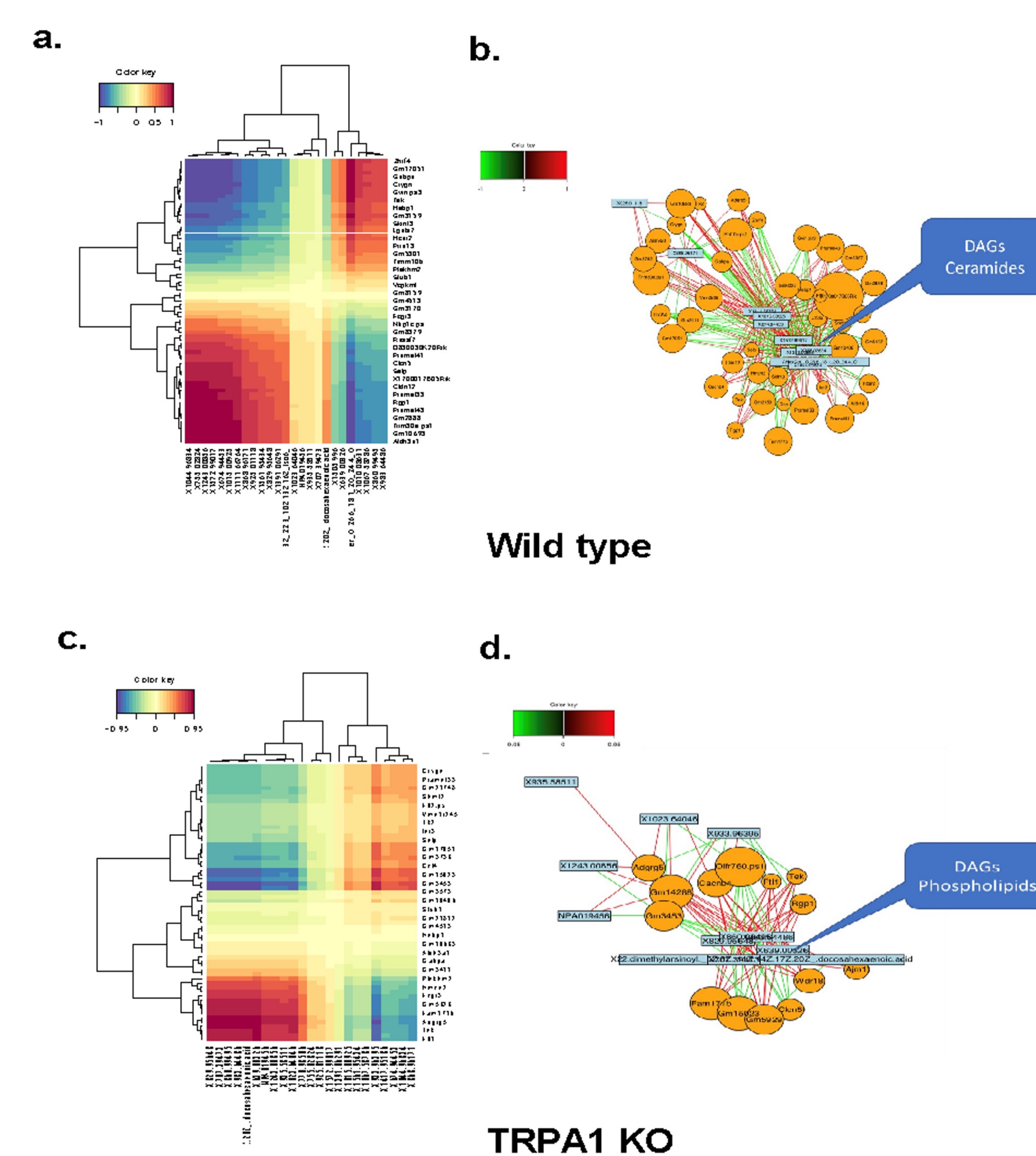


Fig.6: Integrative heatmap: showing the relationship between differential expressed genes and metabolites in Wt and TRPA1KO mice (a and c).Network plot showing the relationship between genes and metabolites and red and green lines represents positive and negative correlations respectively in Wt and TRPA1KO mice respectively(b and d).

Conclusion

- TRPA1 modulation showed much profound effect on infection, by significantly reduced infection severity and abscess size.
- Tissue transcriptomic data coupled with immunohistochemistry revealed that TRPA1 inhibition impacted heat shock protein expression (HSP), which further modulated the HIF-1a and MAPK pathways and reduced IL4 expression.
- Tissue metabolomics reveals that TRP modulation impacts purine and glycosaminoglycan pathways. Multi-omics analysis reveals that diacylglycerol metabolism was the likely bridge between metabolic and immunologic impacts.

Acknowledgements

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